

$$= \begin{cases} 0, & \ell \neq n \\ \frac{NB}{2j} e^{j\psi}, & \ell = n. \end{cases}$$

Dividing these results, we see that with sinusoidal input and output, the frequency response at the frequency $\omega = (2\pi\ell)/NT$ is given by

$$H(e^{j(2\pi\ell)/N}) = \frac{U_\ell}{E_\ell},$$

where $U_\ell = FFT(u_k)$ and $E_\ell = FFT(e_k)$, each evaluated at $n = \ell$. We will discuss in Chapter 12 the general problem of estimation of the total frequency response from experimental data using the DFT/FFT as well as other tools.

4.6 Properties of the z-Transform

We have used the z-transform to show that linear, constant, discrete systems can be described by a transfer function that is the z-transform of the system's unit-pulse response, and we have studied the relationship between the pole-zero patterns of transfer functions in the z-plane and the corresponding time responses. We began a table of z-transforms, and a more extensive table is given in Appendix B. In Section 4.6.1 we turn to consideration of some of the properties of the z-transform that are essential to the effective and correct use of this important tool. In Section 4.6.2 convergence issues concerning the z-transform are discussed and in Section 4.6.3 an alternate derivation of the transfer function is given.

4.6.1 Essential Properties

In order to make maximum use of a table of z-transforms, one must be able to use a few simple properties of the z-transform which follow directly from the definition. Some of these, such as linearity, we have already used without making a formal statement of it, and others, such as the transform of the convolution, we have previously derived. For reference, we will demonstrate a few properties here and collect them into Appendix B for future reference. In all the properties listed below, we assume that $F_i(z) = \mathcal{Z}\{f_i(kT)\}$.

1. **Linearity:** A function $f(x)$ is linear if $f(\alpha x_1 + \beta x_2) = \alpha f(x_1) + \beta f(x_2)$. Applying this result to the definition of the z-transform, we find immediately that

$$\begin{aligned} \mathcal{Z}\{\alpha f_1(kT) + \beta f_2(kT)\} &= \sum_{k=-\infty}^{\infty} \{\alpha f_1(k) + \beta f_2(k)\} z^{-k} \\ &= \alpha \mathcal{Z}\{f_1(k)\} + \beta \mathcal{Z}\{f_2(k)\} \\ &= \alpha F_1(z) + \beta F_2(z). \end{aligned}$$

Thus the z -transform is a linear function. It is the linearity of the transform that makes the partial-fraction technique work.

2. *Convolution of Time Sequences:*

$$\mathcal{Z} \left\{ \sum_{l=-\infty}^{\infty} f_1(l) f_2(k-l) \right\} = F_1(z) F_2(z).$$

We have already developed this result in connection with Eq. (4.32). It is this result with linearity that makes the transform so useful in linear-constant-system analysis because the analysis of a combination of such dynamic systems can be done by linear algebra on the transfer functions.

3. *Time Shift:*

$$\mathcal{Z}\{f(k+n)\} = z^n F(z). \quad (4.113)$$

We demonstrate this result also by direct calculation:

$$\mathcal{Z}\{f(k+n)\} = \sum_{k=-\infty}^{\infty} f(k+n) z^{-k}.$$

If we let $k+n = j$, then

$$\begin{aligned} \mathcal{Z}\{f(k+n)\} &= \sum_{j=-\infty}^{\infty} f(j) z^{-(j-n)} \\ &= z^n F(z). \quad \text{QED} \end{aligned}$$

This property is the essential tool in solving linear constant-coefficient difference equations by transforms. We should note here that the transform of the time shift is not the same for the one-sided transform because a shift can introduce terms with negative argument which are not included in the one-sided transform and must be treated separately. This effect causes initial conditions for the difference equation to be introduced when solution is done with the one-sided transform. See Problem 4.13.

4. *Scaling in the z -Plane:*

$$\mathcal{Z}\{r^{-k} f(k)\} = F(rz). \quad (4.114)$$

By direct substitution, we obtain

$$\begin{aligned} \mathcal{Z}\{r^{-k} f(k)\} &= \sum_{k=-\infty}^{\infty} r^{-k} f(k) z^{-k} \\ &= \sum_{k=-\infty}^{\infty} f(k) (rz)^{-k} \\ &= F(rz). \quad \text{QED} \end{aligned}$$

As an illustration of this property, we consider the z -transform of the unit step, $1(k)$, which we have computed before

$$\mathcal{Z}\{1(k)\} = \sum_{k=0}^{\infty} z^{-k} = \frac{z}{z-1}.$$

By property 4 we have immediately that

$$\mathcal{Z}\{r^{-k}1(k)\} = \frac{rz}{rz-1} = \frac{z}{z-(1/r)}.$$

As a more general example, if we have a polynomial $a(z) = z^2 + a_1z + a_2$ with roots $re^{\pm j\theta}$, then the scaled polynomial $\alpha^2 z^2 + a_1\alpha z + a_2$ has roots $(r/\alpha)e^{\pm j\theta}$. This is an example of radial projection whereby the roots of a polynomial can be projected radially simply by changing the coefficients of the polynomial. The technique is sometimes used in pole-placement designs as described in Chapter 8, and sometimes used in adaptive control as described in Chapter 13.

5. *Final-Value Theorem:* If $F(z)$ converges for $|z| > 1$ and all poles of $(z-1)F(z)$ are inside the unit circle, then

$$\lim_{k \rightarrow \infty} f(k) = \lim_{z \rightarrow 1} (z-1)F(z). \quad (4.115)$$

The conditions on $F(z)$ assure that the only possible pole of $F(z)$ not strictly inside the unit circle is a simple pole at $z = 1$, which is removed in $(z-1)F(z)$. Furthermore, the fact that $F(z)$ converges as the magnitude of z gets arbitrarily large ensures that $f(k)$ is zero for negative k . Therefore, all components of $f(k)$ tend to zero as k gets large, with the possible exception of the constant term due to the pole at $z = 1$. The size of this constant is given by the coefficient of $1/(z-1)$ in the partial-fraction expansion of $F(z)$, namely

$$C = \lim_{z \rightarrow 1} (z-1)F(z).$$

However, because all other terms in $f(k)$ tend to zero, the constant C is the final value of $f(k)$, and Eq. (4.115) results. QED

As an illustration of this property, we consider the signal whose transform is given by

$$U(z) = \frac{z}{z-0.5} \frac{Tz+1}{2z-1}, \quad |z| > 1.$$

Because $U(z)$ satisfies the conditions of Eq. (4.115), we have

$$\begin{aligned} \lim_{k \rightarrow \infty} u(k) &= \lim_{z \rightarrow 1} (z-1) \frac{z}{z-0.5} \frac{Tz+1}{2z-1} \\ &= \lim_{z \rightarrow 1} \frac{z}{z-0.5} \frac{T}{2} (z+1) \end{aligned}$$

$$\begin{aligned}
&= \frac{1}{1 - 0.5} \frac{T}{2} (1 + 1) \\
&= 2T.
\end{aligned}$$

This result can be checked against the closed form for $u(k)$ given by Eq. (4.121) below.

6. *Inversion:* As with the Laplace transform, the z -transform is actually one of a pair of transforms that connect functions of time to functions of the complex variable z . The z -transform computes a function of z from a sequence in k . (We identify the sequence number k with time in our analysis of dynamic systems, but there is nothing in the transform *per se* that requires this.) The inverse z -transform is a means to compute a sequence in k from a given function of z . We first examine two elementary schemes for inversion of a given $F(z)$ which can be used if we know beforehand that $F(z)$ is rational in z and converges as z approaches infinity. For a sequence $f(k)$, the z -transform has been defined as

$$F(z) = \sum_{k=-\infty}^{\infty} f(k)z^{-k}, \quad r_0 < |z| < R_0. \quad (4.116)$$

If any value of $f(k)$ for negative k is nonzero, then there will be a term in Eq. (4.116) with a positive power of z . This term will be unbounded if the magnitude of z is unbounded; and thus if $F(z)$ converges as $|z|$ approaches infinity, we know that $f(k)$ is zero for $k < 0$. In this case, Eq. (4.116) is one-sided, and we can write

$$F(z) = \sum_{k=0}^{\infty} f(k)z^{-k}, \quad r_0 < |z|. \quad (4.117)$$

The right-hand side of Eq. (4.117) is a series expansion of $F(z)$ about infinity or about $z^{-1} = 0$. Such an expansion is especially easy if $F(z)$ is the ratio of two polynomials in z^{-1} . We need only divide the numerator by the denominator in the correct way, and when the division is done, the coefficient of z^{-k} is automatically the sequence value $f(k)$. An example we have worked out before will illustrate the process.

◆ Example 4.12 z -Transform Inversion by Long Division

The system for trapezoid-rule integration has the transfer function given by Eq. (4.14)

$$H(z) = \frac{T}{2} \frac{z+1}{z-1}, \quad |z| > 1.$$

Determine the output for an input which is the geometric series represented by $e_3(k)$ with $r = 0.5$. That is

$$E_3(z) = \frac{z}{z-0.5}, \quad |z| > 0.5.$$

Solution. The z-transform of the output is

$$\begin{aligned} U(z) &= E_3(z)H(z) \\ &= \frac{z}{z-0.5} \frac{T}{2} \frac{z+1}{z-1}, \quad |z| > 1. \end{aligned} \quad (4.118)$$

Equation (4.118) represents the transform of the system output, $u(k)$. Keeping out the factor of $T/2$, we write $U(z)$ as a ratio of polynomials in z^{-1}

$$U(z) = \frac{T}{2} \frac{1+z^{-1}}{1-1.5z^{-1}+0.5z^{-2}}, \quad (4.119)$$

and divide as follows

$$\begin{array}{r} 1 - 1.5z^{-1} + 0.5z^{-2} \overline{) \frac{T}{2} [1 + 2.5z^{-1} + 3.25z^{-2} + 3.625z^{-3} + \dots]} \\ \underline{1 + z^{-1}} \phantom{+ 3.25z^{-2} + 3.625z^{-3} + \dots} \\ 2.5z^{-1} - 0.5z^{-2} \phantom{+ 3.625z^{-3} + \dots} \\ \underline{2.5z^{-1} - 3.75z^{-2} + 1.25z^{-3}} \\ 3.25z^{-2} - 1.25z^{-3} \\ \underline{3.25z^{-2} - 4.875z^{-3} + 1.625z^{-4}} \\ 3.625z^{-3} - 1.625z^{-4} \\ \underline{3.625z^{-3} - \dots} \end{array}$$

By direct comparison with $U(z) = \sum_{k=0}^{\infty} u(k)z^{-k}$, we conclude that

$$\begin{aligned} u_0 &= T/2, \\ u_1 &= (T/2)2.5, \\ u_2 &= (T/2)3.25, \\ &\vdots \end{aligned} \quad (4.120)$$

Clearly, the use of a computer will greatly aid the speed of this process in all but the simplest of cases. Some may prefer to use synthetic division and omit copying over all the extraneous z 's in the division. The process is identical to converting $F(z)$ to the equivalent difference equation and solving for the unit-pulse response.

The second special method for the inversion of z-transforms is to decompose $F(z)$ by partial-fraction expansion and look up the components of the sequence $f(k)$ in a previously prepared table.

◆ Example 4.13 z-Transform Inversion by Partial Fraction Expansion

Repeat Example 4.12 using the partial fraction expansion method.

Solution. Consider again Eq. (4.118) and expand $U(z)$ as a function of z^{-1} as follows

$$U(z) = \frac{T}{2} \frac{1+z^{-1}}{1-z^{-1}} \frac{1}{1-0.5z^{-1}} = \frac{A}{1-z^{-1}} + \frac{B}{1-0.5z^{-1}}.$$

We multiply both sides by $1-z^{-1}$, let $z^{-1} = 1$, and compute

$$A = \frac{T}{2} \frac{2}{0.5} = 2T.$$

Similarly, at $z^{-1} = 2$, we evaluate

$$B = \frac{T}{2} \frac{1+2}{1-2} = -\frac{3T}{2}.$$

Looking back now at e_2 and e_3 , which constitute our “table” for the moment, we can copy down that

$$\begin{aligned} u_k &= Ae_2(k) + Be_3(k) \\ &= 2Te_2(k) - \frac{3T}{2}e_3(k) \\ &= \left(2T - \frac{3T}{2} \left(\frac{1}{2}\right)^k\right) 1(k) \\ &= \frac{T}{2} \left[4 - \frac{3}{2^k}\right] 1(k). \end{aligned} \quad (4.121)$$

Evaluation of Eq. (4.121) for $k = 0, 1, 2, \dots$ will, naturally, give the same values for $u(k)$ as we found in Eq. (4.120).

4.6.2 *Convergence of z-Transform

We now examine more closely the role of the region of convergence of the z-transform and present the inverse-transform integral. We begin with another example. The sequence

$$f(k) = \begin{cases} -1 & k < 0 \\ 0 & k \geq 0 \end{cases}$$

has the transform

$$\begin{aligned} F(z) &= \sum_{k=-\infty}^{-1} -z^{-k} = - \left[\sum_0^{\infty} z^k - 1 \right] \\ &= \frac{z}{z-1}, \quad |z| < 1. \end{aligned}$$

This transform is exactly the same as the transform of the unit step $1(k)$, Eq. (4.95), except that this transform converges *inside* the unit circle and the transform of the $1(k)$ converges outside the unit circle. Knowledge of the region of convergence

is obviously essential to the proper inversion of the transform to obtain the time sequence. The inverse z -transform is the closed, complex integral²²

$$f(k) = \frac{1}{2\pi j} \oint F(z) z^k \frac{dz}{z}, \quad (4.122)$$

where the contour is a circle in the region of convergence of $F(z)$. To demonstrate the correctness of the integral and to use it to compute inverses, it is useful to apply Cauchy's residue calculus [see Churchill and Brown (1984)]. Cauchy's result is that a closed integral of a function of z which is analytic on and inside a closed contour C except at a finite number of isolated singularities z_i is given by

$$\frac{1}{2\pi j} \oint_C F(z) dz = \sum_i \text{Res}(z_i). \quad (4.123)$$

In Eq. (4.123), $\text{Res}(z_i)$ means the residue of $F(z)$ at the singularity at z_i . We will be considering only rational functions, and these have only poles as singularities. If $F(z)$ has a pole of order n at z_1 , then $(z - z_1)^n F(z)$ is regular at z_1 and can be expanded in a Taylor series near z_1 as

$$(z - z_1)^n F(z) = A_{-n} + A_{-n+1}(z - z_1) + \cdots + A_{-1}(z - z_1)^{n-1} + A_0(z - z_1)^n + \cdots. \quad (4.124)$$

The residue of $F(z)$ at z_1 is A_{-1} .

First we will use Cauchy's formula to verify Eq. (4.123). If $F(z)$ is the z -transform of $f(k)$, then we write

$$\mathcal{I} = \frac{1}{2\pi j} \oint \sum_{l=-\infty}^{\infty} f(l) z^{-l} z^k \frac{dz}{z}.$$

We assume that the series for $F(z)$ converges uniformly on the contour of integration, so the series can be integrated term by term. Thus we have

$$\mathcal{I} = \frac{1}{2\pi j} \sum_{l=-\infty}^{\infty} f(l) \oint z^{k-l} \frac{dz}{z}.$$

The argument of the integral has no pole inside the contour if $k - l \geq 1$, and it has zero residue at the pole at $z = 0$ if $k - l < 0$. Only if $k = l$ does the integral have a residue, and that is 1. By Eq. (4.123), the integral is zero if $k \neq l$ and is $2\pi j$ if $k = l$. Thus $\mathcal{I} = f(k)$, which demonstrates Eq. (4.122).

²² If it is known that $f(k)$ is causal, that is, $f(k) = 0$ for $k < 0$, then the region of convergence is outside the smallest circle that contains all the poles of $F(z)$ for rational transforms. It is this property that permits inversion by partial-fraction expansion and long division.

To illustrate the use of Eq. (4.123) to compute the inverse of a z -transform, we will use the function $z/(z-1)$ and consider first the case of convergence for $|z| > 1$ and second the case of convergence for $|z| < 1$. For the first case

$$f_1(k) = \frac{1}{2\pi j} \oint_{|z|=R>1} \frac{z}{z-1} z^k \frac{dz}{z}, \quad (4.125)$$

where the contour is a circle of radius greater than 1. Suppose $k < 0$. In this case, the argument of the integral has two poles inside the contour: one at $z = 1$ with residue

$$\lim_{z \rightarrow 1} (z-1) \frac{z^k}{z-1} = 1,$$

and one pole at $z = 0$ with residue found as in (2.109) (if $k < 0$, then z^{-k} removes the pole)

$$\begin{aligned} z^{-k} \frac{z^k}{z-1} &= \frac{-1}{z-1} \\ &= -(1 + z^{-1} + z^{-2} + \cdots + z^{-k} + \cdots). \end{aligned}$$

The residue is thus -1 for all k , and the sum of the residues is zero, and

$$f_1(k) = 0, \quad k < 0. \quad (4.126)$$

For $k \geq 0$, the argument of the integral in Eq. (4.125) has only the pole at $z = 1$ with residue 1. Thus

$$f_1(k) = 1, \quad k \geq 0. \quad (4.127)$$

Equations (4.123) and (4.124) correspond to the unit-step function, as they should. We would write the inverse transform symbolically $\mathcal{Z}^{-1}\{.\}$ as, in this case

$$\mathcal{Z}^{-1} \left\{ \frac{z}{z-1} \right\} = 1(k) \quad (4.128)$$

when $z/(z-1)$ converges for $|z| > 1$.

If, on the other hand, convergence is inside the unit circle, then for $k \geq 0$, there are no poles of the integrand contained in the contour, and

$$f_2(k) = 0, \quad k \geq 0.$$

At $k < 0$, there is a pole at the origin of z , and as before, the residue is equal to -1 there, so

$$f_2(k) = -1, \quad k < 0.$$

In symbols, corresponding to Eq. (4.128), we have

$$\mathcal{Z}^{-1} \left\{ \frac{z}{z-1} \right\} = 1(k) - 1$$

when $z/(z-1)$ converges for $|z| < 1$.

Although, as we have just seen, the inverse integral can be used to compute an expression for a sequence to which a transform corresponds, a more effective use of the integral is in more general manipulations. We consider one such case that will be of some interest later. First, we consider an expression for the transform of a product of two sequences. Suppose we have

$$f_3(k) = f_1(k)f_2(k),$$

and f_1 and f_2 are such that the transform of the product exists. An expression for $F_3(z)$ in terms of $F_1(z)$ and $F_2(z)$ can be developed as follows. By definition

$$F_3(z) = \sum_{k=-\infty}^{\infty} f_1(k)f_2(k)z^{-k}.$$

From the inversion integral, Eq. (4.122), we can replace $f_2(k)$ by an integral

$$F_3(z) = \sum_{k=-\infty}^{\infty} f_1(k)z^{-k} \frac{1}{2\pi j} \oint_{C_2} F_2(\zeta)\zeta^k \frac{d\zeta}{\zeta}.$$

We assume that we can find a region where we can exchange the summation with the integration. The contour will be called C_3 in this case

$$F_3(z) = \frac{1}{2\pi j} \oint_{C_3} F_2(\zeta) \sum_{k=-\infty}^{\infty} f_1(k) \left(\frac{z}{\zeta}\right)^{-k} \frac{d\zeta}{\zeta}.$$

The sum can now be recognized as $F_1(z/\zeta)$ and, when we substitute this,

$$F_3(z) = \frac{1}{2\pi j} \oint_{C_3} F_2(\zeta) F_1\left(\frac{z}{\zeta}\right) \frac{d\zeta}{\zeta}, \quad (4.129)$$

the contour C_3 must be in the overlap of the convergence regions of $F_2(\zeta)$ and $F_1(z/\zeta)$. Then $F_3(z)$ will converge for the range of values of z for which C_3 can be found.

If we let $f_1 = f_2$ and $z = 1$ in Eq. (4.129), we have the discrete version of Parseval's theorem, where convergence is on the unit circle

$$F_3(1) = \sum_{k=-\infty}^{\infty} f_1^2 = \frac{1}{2\pi j} \oint_C F_1(\zeta) F_1\left(\frac{1}{\zeta}\right) \frac{d\zeta}{\zeta}. \quad (4.130)$$

This particular theorem shows how we can compute the sum of squares of a time sequence by evaluating a complex integral in the z -domain. The result is useful in the design of systems by least squares.

4.6.3 *Another Derivation of the Transfer Function

Let $\mathcal{D}$ be a discrete system which maps an input sequence, $\{e(k)\}$, into an output sequence $\{u(k)\}$.²³ Then, expressing this as an operator on $e(k)$, we have

$$u(k) = \mathcal{D}\{e(k)\}.$$

If $\mathcal{D}$ is linear, then

$$\mathcal{D}\{\alpha e_1(k) + \beta e_2(k)\} = \alpha \mathcal{D}\{e_1(k)\} + \beta \mathcal{D}\{e_2(k)\}. \quad (4.131)$$

If the system is time invariant, a shift in $e(k)$ to $e(k + j)$ must result in no other effects but a shift in the response, u . We write

$$\mathcal{D}\{e(k + j)\} = u(k + j) \quad \text{for all } j \quad (4.132)$$

if

$$\mathcal{D}\{e(k)\} = u(k).$$

Theorem

If $\mathcal{D}$ is linear and time invariant and is given an input z^k for a value of z for which the output is finite at time k , then the output will be of the form $H(z)z^k$.

In general, if $e(k) = z^k$, then an arbitrary finite response can be written

$$u(k) = H(z, k)z^k.$$

Consider $e_2(k) = z^{k+j} = z^j z^k$ for some fixed j . From Eq. (4.131), if we let $\alpha = z^j$, it must follow that

$$\begin{aligned} u_2 &= z^j u(k) \\ &= z^j H(z, k)z^k \\ &= H(z, k)z^{k+j}. \end{aligned} \quad (4.133)$$

From Eq. (4.132), we must have

$$\begin{aligned} u_2(k) &= u(k + j) \\ &= H(z, j + k)z^{k+j} \quad \text{for all } j. \end{aligned} \quad (4.134)$$

From a comparison of Eqs. (4.133) and (4.134), it follows that

$$H(z, k) = H(z, k + j) \quad \text{for all } j$$

that is, H does not depend on the second argument and can be written $H(z)$. Thus for the elemental signal $e(k) = z^k$, we have a solution $u(k)$ of the same (exponential) shape but modulated by a ratio $H(z)$, $u(k) = H(z)z^k$.

²³ This derivation was suggested by L. A. Zadeh in 1952 at Columbia University.

Can we represent a general signal as a *linear sum* (integral) of such elements? We can, by the inverse integral derived above, as follows

$$e(k) = \frac{1}{2\pi j} \oint E(z) z^k \frac{dz}{z}, \quad (4.135)$$

where

$$E(z) = \sum_{-\infty}^{\infty} e(k) z^{-k}, \quad r < |z| < R, \quad (4.136)$$

for signals with $r < R$ for which Eq. (4.136) converges. We call $E(z)$ the z-transform of $e(k)$, and the (closed) path of integration is in the annular region of convergence of Eq. (4.136). If $e(k) = 0$, $k < 0$, then $R \rightarrow \infty$, and this region is the whole z-plane *outside* a circle of finite radius.

The consequences of linearity are that the response to a sum of signals is the sum of the responses as given in Eq. (4.131). Although Eq. (4.135) is the limit of a sum, the result still holds, and we can write

$$u(k) = \frac{1}{2\pi j} \oint E(z) [\text{response to } z^k] \frac{dz}{z},$$

but, by the theorem, the response to z^k is $H(z)z^k$. Therefore we can write

$$\begin{aligned} u(k) &= \frac{1}{2\pi j} \oint E(z) [H(z)z^k] \frac{dz}{z} \\ &= \frac{1}{2\pi j} \oint H(z) E(z) z^k \frac{dz}{z}. \end{aligned} \quad (4.137)$$

We can define $U(z) = H(z)E(z)$ by comparison with Eq. (4.135) and note that

$$U(z) = \sum_{k=-\infty}^{\infty} u(k) z^{-k} = H(z)E(z). \quad (4.138)$$

Thus $H(z)$ is the *transfer function*, which is the ratio of the transforms of $e(k)$ and $u(k)$ as well as the amplitude response to inputs of the form z^k .

This derivation begins with linearity and stationarity and derives the z-transform as the natural tool of analysis from the fact that input signals in the form z^k produce an output that has the same shape.²⁴ It is somewhat more satisfying to derive the necessary transform than to start with the transform and see what systems it is good for. Better to start with the problem and find a tool than start with a tool and look for a problem. Unfortunately, the direct approach requires extensive use of the inversion integral and more sophisticated analysis to develop the main result, which is Eq. (4.138). *Chacun à son goût.*

²⁴ Because z^k is unchanged in shape by passage through the linear constant system, we say that z^k is an eigenfunction of such systems.

4.7 Summary

- The z -transform can be used to solve discrete difference equations in the same way that the Laplace transform is used to solve continuous differential equations.
- The key property of the z -transform that allows solution of difference equations is

$$\mathcal{Z}\{f(k-1)\} = z^{-1}F(z). \quad (4.113)$$

- A system will be stable in the sense that a *Bounded Input* will yield a *Bounded Output* (BIBO stability) if

$$\sum_{l=-\infty}^{\infty} |h_{k-l}| < \infty. \quad (4.35)$$

- A discrete system can be defined by its transfer function (in z) or its state-space difference equation.
- The z -transform of the samples of a continuous system $G(s)$ preceded by a zero-order-hold (ZOH) is

$$G(z) = (1 - z^{-1})\mathcal{Z}\left\{\frac{G(s)}{s}\right\} \quad (4.41)$$

which is typically evaluated using MATLAB's `c2d.m`.

- For the continuous state-space model

$$\dot{\mathbf{x}} = \mathbf{F}\mathbf{x} + \mathbf{G}u, \quad (4.45)$$

$$y = \mathbf{H}\mathbf{x} + Ju, \quad (4.46)$$

preceded by a zero-order-hold, the discrete state-space difference equations are

$$\begin{aligned} \mathbf{x}(k+1) &= \Phi\mathbf{x}(k) + \Gamma u(k), \\ y(k) &= \mathbf{H}\mathbf{x}(k) + Ju(k), \end{aligned} \quad (4.59)$$

where

$$\begin{aligned} \Phi &= e^{\mathbf{F}T} \\ \Gamma &= \int_0^T e^{\mathbf{F}\eta} d\eta \mathbf{G}, \end{aligned} \quad (4.58)$$

which can be evaluated by MATLAB's `c2d.m`.

- The discrete transfer function in terms of the state-space matrices is

$$\frac{Y(z)}{U(z)} = \mathbf{H}[z\mathbf{I} - \Phi]^{-1}\Gamma, \quad (4.64)$$

which can be evaluated in MATLAB by the `tf` function.

- The characteristic behavior associated with poles in the z -plane is shown in Figs. 4.21 through 4.23 and summarized in Fig. 4.25. Responses are typically determined via MATLAB's `impz` or `step`.
- A system represented by $H(z)$ has a discrete frequency response to sinusoids at ω_o , given by an amplitude, A , and phase, ψ , as

$$A = |H(e^{j\omega_o T})| \quad \text{and} \quad \psi = \angle(H(e^{j\omega_o T})) \quad (4.110)$$

which can be evaluated by MATLAB's `bode`.

- The discrete Final Value Theorem, for an $F(z)$ that converges and has a final value, is given by

$$\lim_{k \rightarrow \infty} f(k) = \lim_{z \rightarrow 1} (z - 1) F(z). \quad (4.115)$$

4.8 Problems

4.1 Check the following for stability:

- (a) $u(k) = 0.5u(k-1) - 0.3u(k-2)$
- (b) $u(k) = 1.6u(k-1) - u(k-2)$
- (c) $u(k) = 0.8u(k-1) + 0.4u(k-2)$

4.2 (a) Derive the difference equation corresponding to the approximation of integration found by fitting a parabola to the points e_{k-2} , e_{k-1} , e_k and taking the area under this parabola between $t = kT - T$ and $t = kT$ as the approximation to the integral of $e(t)$ over this range.

- (b) Find the transfer function of the resulting discrete system and plot the poles and zeros in the z -plane.

4.3 Verify that the transfer function of the system of Fig. 4.8(c) is given by the same $H(z)$ as the system of Fig. 4.9(c).

- 4.4 (a) Compute and plot the unit-pulse response of the system derived in Problem 4.2.
- (b) Is this system BIBO stable?

4.5 Consider the difference equation

$$u(k+2) = 0.25u(k).$$

- (a) Assume a solution $u(k) = A_1 z_1^k$ and find the characteristic equation in z .
- (b) Find the characteristic roots z_1 and z_2 and decide if the equation solutions are stable or unstable.
- (c) Assume a general solution of the form

$$u(k) = A_1 z_1^k + A_2 z_2^k$$

and find A_1 and A_2 to match the initial conditions $u(0) = 0$, $u(1) = 1$.

- (d) Repeat parts (a), (b), and (c) for the equation

$$u(k+2) = -0.25u(k).$$

- (e) Repeat parts (a), (b), and (c) for the equation

$$u(k+2) = u(k+1) - 0.5u(k).$$

- 4.6 Show that the characteristic equation

$$z^2 - 2r \cos(\theta)z + r^2$$

has the roots

$$z_{1,2} = re^{\pm j\theta}.$$

- 4.7 (a) Use the method of block-diagram reduction, applying Figs. 4.5, 4.6, and 4.7 to compute the transfer function of Fig. 4.8(c).
 (b) Repeat part (a) for the diagram of Fig. 4.9(c).

- 4.8 Use MATLAB to determine how many roots of the following are outside the unit circle.

- (a) $z^2 + 0.25 = 0$
 (b) $z^3 - 1.1z^2 + 0.01z + 0.405 = 0$
 (c) $z^3 - 3.6z^2 + 4z - 1.6 = 0$

- 4.9 Compute by hand and table look-up the discrete transfer function if the
- $G(s)$
- in Fig. 4.12 is

- (a) $\frac{K}{s}$
 (b) $\frac{3}{s(s+3)}$
 (c) $\frac{3}{(s+1)(s+3)}$
 (d) $\frac{(s+1)}{s^2}$
 (e) $\frac{e^{sT/2}}{s^2}$
 (f) $\frac{(1-s)}{s^2}$
 (g) $\frac{3e^{-1.5Ts}}{(s+1)(s+3)}$

- (h) Repeat the calculation of these discrete transfer functions using MATLAB. Compute for the sampling period $T = 0.05$ and $T = 0.5$ and plot the location of the poles and zeros in the z -plane.

- 4.10 Use MATLAB to compute the discrete transfer function if the
- $G(s)$
- in Fig. 4.12 is

- (a) the two-mass system with the non-colocated actuator and sensor of Eq. (A.21) with sampling periods $T = 0.02$ and $T = 0.1$. Plot the zeros and poles of the results in the z -plane. Let $\omega_p = 5$, $\zeta_p = 0.01$.
 (b) the two-mass system with the colocated actuator and sensor given by Eq. (A.23). Use $T = 0.02$ and $T = 0.1$. Plot the zeros and poles of the results in the z -plane. Let $\omega_p = 5$, $\omega_z = 3$, $\zeta_p = \zeta_z = 0$.
 (c) the two-input-two-output paper machine described in Eq. (A.24). Let $T = 0.1$ and $T = 0.5$.

- 4.11 Consider the system described by the transfer function

$$\frac{Y(s)}{U(s)} = G(s) = \frac{3}{(s+1)(s+3)}.$$

- (a) Draw the block diagram corresponding to this system in control canonical form, define the state vector, and give the corresponding description matrices **F**, **G**, **H**, **J**.

- (b) Write $G(s)$ in partial fractions and draw the corresponding parallel block diagram with each component part in control canonical form. Define the state ξ and give the corresponding state description matrices $\mathbf{A}$, $\mathbf{B}$, $\mathbf{C}$, $\mathbf{D}$.
- (c) By finding the transfer functions X_1/U and X_2/U of part (a) in partial fraction form, express x_1 and x_2 in terms of ξ_1 and ξ_2 . Write these relations as the two-by-two transformation $\mathbf{T}$ such that $\mathbf{x} = \mathbf{T}\xi$.
- (d) Verify that the matrices you have found are related by the formulas

$$\mathbf{A} = \mathbf{T}^{-1}\mathbf{F}\mathbf{T},$$

$$\mathbf{B} = \mathbf{T}^{-1}\mathbf{G},$$

$$\mathbf{C} = \mathbf{H}\mathbf{T},$$

$$\mathbf{D} = \mathbf{J}.$$

4.12 The first-order system $(z - \alpha)/(1 - \alpha)z$ has a zero at $z = \alpha$.

- (a) Plot the step response for this system for $\alpha = 0.8, 0.9, 1.1, 1.2, 2$.
- (b) Plot the overshoot of this system on the same coordinates as those appearing in Fig. 4.30 for $-1 < \alpha < 1$.
- (c) In what way is the step response of this system unusual for $\alpha > 1$?

4.13 The one-sided z -transform is defined as

$$F(z) = \sum_{k=0}^{\infty} f(k)z^{-k}.$$

- (a) Show that the one-sided transform of $f(k+1)$ is

$$\mathcal{Z}\{f(k+1)\} = zF(z) - zf(0).$$

- (b) Use the one-sided transform to solve for the transforms of the Fibonacci numbers by writing Eq. (4.4) as $u_{k+2} = u_{k+1} + u_k$. Let $u_0 = u_1 = 1$. [You will need to compute the transform of $f(k+2)$.]
- (c) Compute the location of the poles of the transform of the Fibonacci numbers.
- (d) Compute the inverse transform of the numbers.
- (e) Show that if u_k is the k th Fibonacci number, then the ratio u_{k+1}/u_k will go to $(1 + \sqrt{5})/2$, the golden ratio of the Greeks.
- (f) Show that if we add a forcing term, $e(k)$, to Eq. (4.4) we can generate the Fibonacci numbers by a system that can be analyzed by the two-sided transform; i.e., let $u_k = u_{k-1} + u_{k-2} + e_k$ and let $e_k = \delta_0(k)$ ($\delta_0(k) = 1$ at $k = 0$ and zero elsewhere). Take the two-sided transform and show the same $U(z)$ results as in part (b).

4.14 Substitute $u = Az^k$ and $e = Bz^k$ into Eqs. (4.2) and (4.7) and show that the transfer functions, Eqs. (4.15) and (4.14), can be found in this way.

4.15 Consider the transfer function

$$H(z) = \frac{(z+1)(z^2 - 1.3z + 0.81)}{(z^2 - 1.2z + 0.5)(z^2 - 1.4z + 0.81)}.$$

Draw a *cascade* realization, using observer canonical forms for second-order blocks and in such a way that the coefficients as shown in $H(z)$ above are the parameters of the block diagram.

- 4.16 (a) Write the $H(z)$ of Problem 4.15 in partial fractions in two terms of second order each, and draw a *parallel* realization, using the observer canonical form for each block and showing the coefficients of the partial-fraction expansion as the parameters of the realization.
- (b) Suppose the two factors in the denominator of $H(z)$ were identical (say we change the 1.4 to 1.2 and the 0.81 to 0.5). What would the parallel realization be in this case?
- 4.17 Show that the observer canonical form of the system equations shown in Fig. 4.9 can be written in the state-space form as given by Eq. (4.27).
- 4.18 Draw out each block of Fig. 4.10 in (a) control and (b) observer canonical form. Write out the state-description matrices in each case.
- 4.19 For a second-order system with damping ratio 0.5 and poles at an angle in the z -plane of $\theta = 30^\circ$, what percent overshoot to a step would you expect if the system had a zero at $z_2 = 0.6$?
- 4.20 Consider a signal with the transform (which converges for $|z| > 2$)

$$U(z) = \frac{z}{(z-1)(z-2)}.$$

- (a) What value is given by the formula (Final Value Theorem) of (4.115) applied to this $U(z)$?
- (b) Find the final value of $u(k)$ by taking the inverse transform of $U(z)$, using partial-fraction expansion and the tables.
- (c) Explain why the two results of (a) and (b) differ.
- 4.21 (a) Find the z -transform and be sure to give the region of convergence for the signal

$$u(k) = r^{|k|}, \quad |r| < 1.$$

[Hint: Write u as the sum of two functions, one for $k \geq 0$ and one for $k < 0$, find the individual transforms, and determine values of z for which *both* terms converge.]

- (b) If a rational function $U(z)$ is known to converge on the unit circle $|z| = 1$, show how partial-fraction expansion can be used to compute the inverse transform. Apply your result to the transform you found in part (a).
- 4.22 Compute the inverse transform, $f(k)$, for each of the following transforms:
- (a) $F(z) = \frac{1}{1+z^{-2}}, \quad |z| > 1;$
- (b) $F(z) = \frac{z(z-1)}{z^2 - 1.25z + 0.25}, \quad |z| > 1;$
- (c) $F(z) = \frac{z}{z^2 - 2z + 1}, \quad |z| > 1;$
- (d) $F(z) = \frac{z}{(z-\frac{1}{2})(z-2)}, \quad 1/2 < |z| < 2.$
- 4.23 Use MATLAB to plot the time sequence associated with each of the transforms in Problem 4.22.
- 4.24 Use the z -transform to solve the difference equation

$$y(k) - 3y(k-1) + 2y(k-2) = 2u(k-1) - 2u(k-2),$$

$$u(k) = \begin{cases} k, & k \geq 0 \\ 0 & k < 0 \end{cases}$$

$$y(k) = 0, \quad k < 0.$$

4.25 For the difference equation in Problem 4.24, solve using MATLAB.

4.26 Compute by hand and table look-up the discrete transfer function if the $G(s)$ in Fig. 4.12 is

$$G(s) = \frac{10(s+1)}{s^2(s+10)}$$

and the sample period is $T = 10$ msec. Verify the calculation using MATLAB.

4.27 Find the discrete state-space model for the system in Problem 4.26.

4.28 Compute by hand and table look-up the discrete transfer function if the $G(s)$ in Fig. 4.12 is

$$G(s) = \frac{10(s+1)}{s^2 + s + 10}$$

and the sample period is $T = 10$ msec. Verify the calculation using MATLAB and find the DC gain of both the $G(s)$ and the $G(z)$.

4.29 Find the discrete state-space model for the system in Problem 4.28. Then compute the eigenvalues of Φ and the transmission zeros of the state-space model.

4.30 Find the state-space model for Fig. 4.12 with

$$G(s) = \frac{1}{s^2}$$

where there is a one cycle delay after the A/D converter.

• 5 •

Sampled-Data Systems

A Perspective on Sampled-Data Systems

The use of digital logic or digital computers to calculate a control action for a continuous, dynamic system introduces the fundamental operation of sampling. Samples are taken from the continuous physical signals such as position, velocity, or temperature and these samples are used in the computer to calculate the controls to be applied. Systems where discrete signals appear in some places and continuous signals occur in other parts are called *sampled-data systems* because continuous data are sampled before being used. In many ways the analysis of a purely continuous system or of a purely discrete system is simpler than that of sampled-data systems. The analysis of linear, time-invariant continuous systems can be done with the Laplace transform and the analysis of linear time-invariant discrete systems can be done with the z -transform alone. If one is willing to restrict attention to only the samples of all the signals in a digital control one can do much useful analysis and design on the system as a purely discrete system using the z -transform. However the physical reality is that the computer operations are on discrete signals while the plant signals are in the continuous world and in order to consider the behavior of the plant between sampling instants, it is necessary to consider both the discrete actions of the computer and the continuous response of the plant. Thus the role of sampling and the conversion from continuous to discrete and back from discrete to continuous are very important to the understanding of the complete response of digital control, and we must study the process of sampling and how to make mathematical models of analog-to-digital conversion and digital-to-analog conversion. This analysis requires careful treatment using the Fourier transform but the effort is well rewarded with the understanding it provides of sampled-data systems.

Chapter Overview

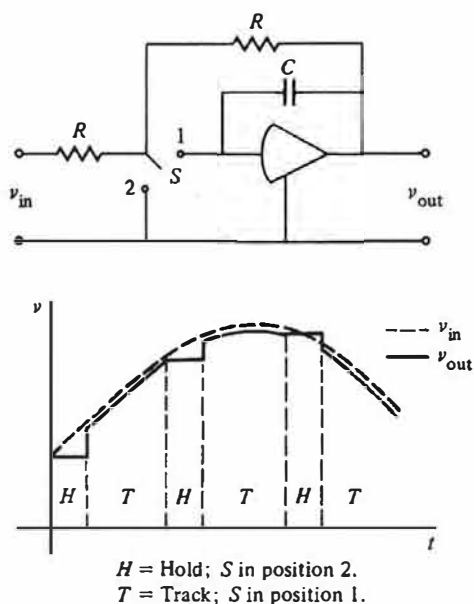
In this chapter, we introduce the analysis of the sampling process and describe both a time-domain and a frequency-domain representation. We also describe the companion process of data extrapolation or data holding to construct a continuous time signal from samples. As part of this analysis we show that a sampled-data system is made time varying by the introduction of sampling, and thus it is not possible to describe such systems exactly by a continuous-time transfer function. However, a continuous signal is recovered by the hold process and we can approximate the sinusoidal response of a sampler and hold by fitting another sinusoid of the same frequency to the complete response. We show how to compute this best-fit sinusoidal response analytically and use it to obtain a good approximation to a transfer function. For those familiar with the idea, this approach is equivalent to the use of the “describing function” that is used to approximate a transfer function for simple nonlinear systems. In Section 5.1 the analysis of the sample and hold operation is considered and in Section 5.2 the frequency analysis of a sampled signal is given. Here the important phenomenon of signal aliasing caused by sampling is introduced. In Section 5.3 the zero-order hold and some of its generalizations are considered. Analysis of sampled-data systems in the frequency domain is introduced in Section 5.4 including block diagram analysis of these combined systems. Finally in Section 5.5 computation of intersample ripple is discussed.

5.1 Analysis of the Sample and Hold

To get samples of a physical signal such as a position or a velocity into digital form, we typically have a sensor that produces a voltage proportional to the physical variable and an **analog-to-digital converter**, commonly called an **A/D converter** or ADC, that transforms the voltage into a digital number. The physical conversion always takes a non-zero time, and in many instances this time is significant with respect to the sample period of the control or with respect to the rate of change of the signal being sampled. In order to give the computer an accurate representation of the signal exactly at the sampling instants kT , the A/D converter is typically preceded by a **sample-and-hold circuit (SHC)**. A simple electronic schematic is sketched in Fig. 5.1, where the switch, S , is an electronic device driven by simple logic from a clock. Its operation is described in the following paragraph.

With the switch, S , in position 1, the amplifier output $v_{\text{out}}(t)$ tracks the input voltage $v_{\text{in}}(t)$ through the transfer function $1/(RCs + 1)$. The circuit bandwidth of the SHC, $1/RC$, is selected to be high compared to the input signal bandwidth. Typical values are $R = 1000$ ohms, $C = 30 \times 10^{-12}$ farads for a bandwidth of $f = 1/2\pi RC = 5.3$ MHz. During this “tracking time,” the ADC is turned off and ignores v_{out} . When a sample is to be taken at $t = kT$ the switch S is set

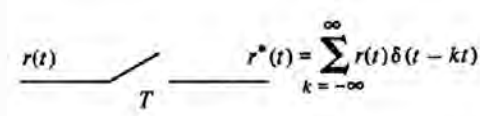
Figure 5.1
Analog-to-digital
converter with sample
and hold



to position 2 and the capacitor C holds the output of the operational amplifier frozen from that time at $v_{out}(kT) = v_{in}(kT)$. The ADC is now signaled to begin conversion of the constant input from the SHC into a digital number which will be a true representation of the input voltage at the sample instant. When the conversion is completed, the digital number is presented to the computer at which time the calculations based on this sample value can begin. The SHC switch is now moved to position 1, and the circuit is again tracking, waiting for the next command to freeze a sample. The SHC needs only to hold the voltage for a short time on the order of microseconds in order for the conversion to be completed before it starts tracking again. The value converted is held inside the computer for the entire sampling period of the system, so the combination of the electronic SHC plus the ADC operate as a sample-and-hold for the sampling period, T , which may be many milliseconds long. The number obtained by the ADC is a quantized version of the signal represented in a finite number of bits, 12 being a typical number. As a result, the device is nonlinear. However, the signals are typically large with respect to the smallest quantum and the effect of this nonlinearity can be ignored in a first analysis. A detailed study of quantization is included in Chapter 10.

For the purpose of the analysis, we separate the sample and hold into two mathematical operations: a sampling operation represented by impulse modulation and a hold operation represented as a linear filter. The symbol or schematic of the ideal sampler is shown in Fig. 5.2; its role is to give a mathematical representation of the process of taking periodic samples from $r(t)$ to produce $r(kT)$ and

Figure 5.2
The sampler



to do this in such a way that we can include the sampled signals in the analysis of continuous signals using the Laplace transform.¹ The technique is to use *impulse modulation* as the mathematical representation of sampling. Thus, from Fig. 5.2, we picture the output of the sampler as a string of impulses

$$r^*(t) = \sum_{k=-\infty}^{\infty} r(t) \delta(t - kT). \quad (5.1)$$

The impulse can be visualized as the limit of a pulse of unit area that has growing amplitude and shrinking duration. The essential property of the impulse is the sifting property that

$$\int_{-\infty}^{\infty} f(t) \delta(t - a) dt = f(a) \quad (5.2)$$

for all functions f that are continuous at a . The integral of the impulse is the unit step

$$\int_{-\infty}^t \delta(\tau) d\tau = 1(t), \quad (5.3)$$

and the Laplace transform of the unit impulse is 1, because

$$\mathcal{L}\{\delta(t)\} = \int_{-\infty}^{\infty} \delta(\tau) e^{-s\tau} d\tau = 1. \quad (5.4)$$

Using these properties we can see that $r^*(t)$, defined in Eq. (5.1), depends only on the discrete sample values $r(kT)$. The Laplace transform of $r^*(t)$ can be computed as follows

$$\mathcal{L}\{r^*(t)\} = \int_{-\infty}^{\infty} r^*(\tau) e^{-s\tau} d\tau.$$

If we substitute Eq. (5.1) for $r^*(t)$, we obtain

$$= \int_{-\infty}^{\infty} \sum_{k=-\infty}^{\infty} r(\tau) \delta(\tau - kT) e^{-s\tau} d\tau,$$

and now, exchanging integration and summation and using Eq. (5.2), we have

$$R^*(s) = \sum_{k=-\infty}^{\infty} r(kT) e^{-skT}. \quad (5.5)$$

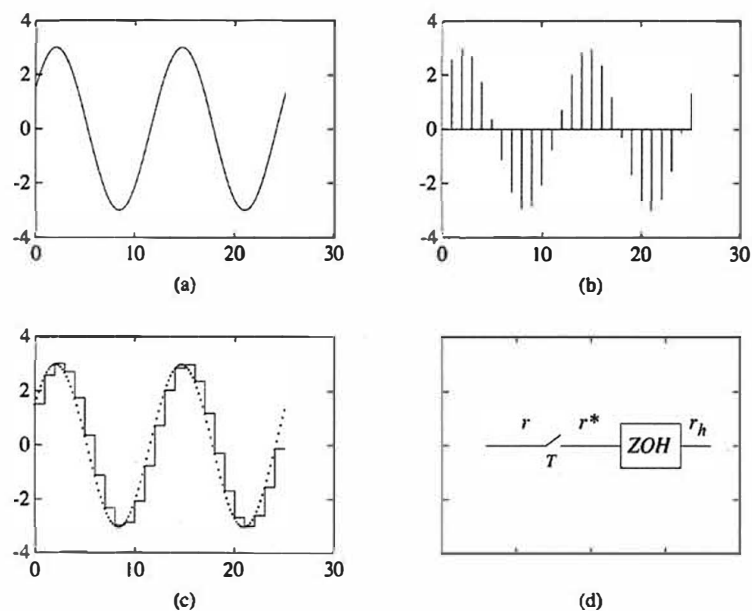
¹ We assume that the reader has some familiarity with Fourier and Laplace transform analysis. A general reference is Bracewell (1978).

The notation $R^*(s)$ is used to symbolize the (Laplace) transform of $r^*(t)$, the sampled or impulse-modulated $r(t)$.² Notice that if the signal $r(t)$ in Eq. (5.1) is shifted a small amount then different samples will be selected by the sampling process for the output proving that sampling is not a time-invariant process. Consequently one must be very careful in using transform analysis in this context.

Having a model of the sampling operation as impulse modulation, we need to model the hold operation to complete the description of the physical sample-and-hold which will take the impulses that are produced by the mathematical sampler and produce the piecewise constant output of the device. Typical signals are sketched in Fig. 5.3. Once the samples are taken, as represented by $r^*(t)$ in

Figure 5.3

The sample and hold, showing typical signals.
(a) Input signal r ;
(b) sampled signal r^* ;
(c) output signal r_h ;
(d) sample and hold



² It will be necessary, from time to time, to consider sampling a signal that is not continuous. The only case we will consider will be equivalent to applying a step function, $1(t)$, to a sampler. For the purposes of this book we will define the unit step to be continuous from the right and assume that the impulse, $\delta(t)$, picks up the full value of unity. By this convention and Eq. (5.1) we compute

$$1^*(t) = \sum_{k=0}^{\infty} \delta(t - kT), \quad (a)$$

and, using Eq. (5.2), we obtain

$$\mathcal{L}\{1^*(t)\} = 1/(1 - e^{-Ts}). \quad (b)$$

The reader should be warned that the Fourier integral converges to the *average* value of a function at a discontinuity and not the value approached from the right as we assume. Because our use of the transform theory is elementary and the convenience of equation (b) above is substantial, we have selected the continuous-from-the-right convention. In case of doubt, the discontinuous term should be separated and treated by special analysis, perhaps in the time domain.

Eq. (5.1), the hold is defined as the means whereby these impulses are extrapolated to the piecewise constant signal $r_h(t)$, defined as

$$r_h(t) = r(kT) \quad kT \leq t < kT + T. \quad (5.6)$$

A general technique of data extrapolation from samples is to use a polynomial fit to the past samples. If the extrapolation is done by a constant, which is a zero-order polynomial, then the extrapolator is called a **zero-order hold**, and its transfer function is designated as $ZOH(s)$. We can compute $ZOH(s)$ as the transform of its impulse response.³ If $r^*(t) = \delta(t)$, then $r_h(t)$, which is now the impulse response of the ZOH , is a pulse of height 1 and duration T seconds. The mathematical representation of the impulse response is simply

$$p(t) = 1(t) - 1(t - T).$$

The required transfer function is the Laplace transform of $p(t)$ as

$$\begin{aligned} ZOH(s) &= \mathcal{L}\{p(t)\} \\ &= \int_0^{\infty} [1(t) - 1(t - T)]e^{-st} dt \\ &= (1 - e^{-sT})/s. \end{aligned} \quad (5.7)$$

Thus the linear behavior of an A/D converter with sample and hold can be modeled by Fig. 5.3. We must emphasize that the impulsive signal $r^*(t)$ in Fig. 5.3 is not expected to represent a physical signal in the A/D converter circuit; rather it is a hypothetical signal introduced to allow us to obtain a transfer-function model of the hold operation and to give an input-output model of the sample-and-hold suitable for transform and other linear systems analysis.

5.2 Spectrum of a Sampled Signal

We can get further insight into the process of sampling by an alternative representation of the transform of $r^*(t)$ using Fourier analysis. From Eq. (5.1) we see that $r^*(t)$ is a product of $r(t)$ and the train of impulses, $\sum \delta(t - kT)$. The latter series, being periodic, can be represented by a Fourier series

$$\sum_{k=-\infty}^{\infty} \delta(t - kT) = \sum_{n=-\infty}^{\infty} C_n e^{j(2\pi n/T)t},$$

where the Fourier coefficients, C_n , are given by the integral over one period as

$$C_n = \frac{1}{T} \int_{-T/2}^{T/2} \sum_{k=-\infty}^{\infty} \delta(t - kT) e^{-jn(2\pi t/T)} dt.$$

³ The hold filter in Fig. 5.3(d) will receive one unit-size impulse if the input signal is zero at every sample time except $t = 0$ and is equal to 1 there. That is, if $r(kT) = 0$, $k \neq 0$ and $r(0) = 1$.

The only term in the sum of impulses that is in the range of the integral is the $\delta(t)$ at the origin, so the integral reduces to

$$C_n = \frac{1}{T} \int_{-T/2}^{T/2} \delta(t) e^{-jn(2\pi t/T)} dt;$$

but the sifting property from Eq. (5.2) makes this easy to integrate, with the result

$$C_n = \frac{1}{T}.$$

Thus we have derived the representation for the sum of impulses as a Fourier series

$$\sum_{k=-\infty}^{\infty} \delta(t - kT) = \frac{1}{T} \sum_{n=-\infty}^{\infty} e^{j(2\pi n/T)t}. \quad (5.8)$$

We define $\omega_s = 2\pi/T$ as the sampling frequency (in radians per second) and now substitute Eq. (5.8) into Eq. (5.1) using ω_s . We take the Laplace transform of the output of the mathematical sampler,

$$\mathcal{L}\{r^*(t)\} = \int_{-\infty}^{\infty} r(t) \left\{ \frac{1}{T} \sum_{n=-\infty}^{\infty} e^{jn\omega_s t} \right\} e^{-st} dt$$

and integrate the sum, term by term to obtain

$$R^*(s) = \frac{1}{T} \sum_{n=-\infty}^{\infty} \int_{-\infty}^{\infty} r(t) e^{jn\omega_s t} e^{-st} dt.$$

If we combine the exponentials in the integral, we get

$$R^*(s) = \frac{1}{T} \sum_{n=-\infty}^{\infty} \int_{-\infty}^{\infty} r(t) e^{-(s - jn\omega_s)t} dt.$$

The integral here is the Laplace transform of $r(t)$ with only a change of variable where the frequency goes. The result can therefore be written as

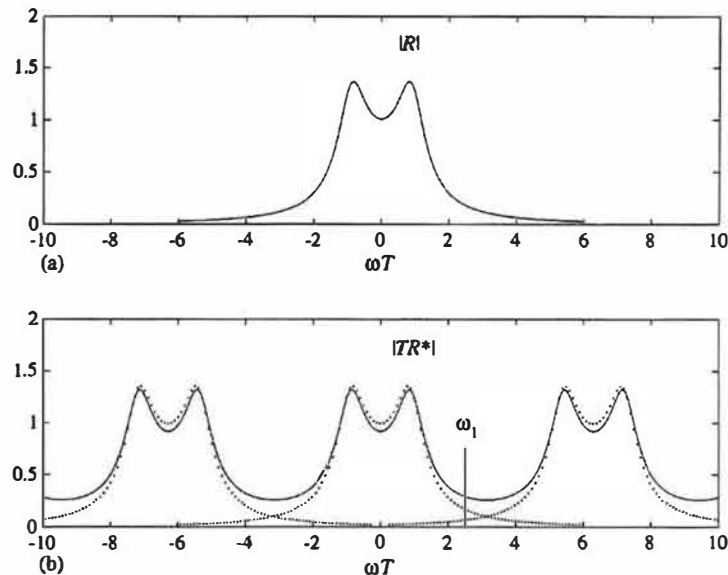
$$R^*(s) = \frac{1}{T} \sum_{n=-\infty}^{\infty} R(s - jn\omega_s), \quad (5.9)$$

where $R(s)$ is the transform of $r(t)$. In communication or radio engineering terms, Eq. (5.8) expresses the fact that the impulse train corresponds to an infinite sequence of carrier frequencies at integral values of $2\pi/T$, and Eq. (5.9) shows that when $r(t)$ modulates all these carriers, it produces a never-ending train of sidebands. A sketch of the elements in the sum given in Eq. (5.9) is shown in Fig. 5.4.

An important feature of sampling, shown in Fig. 5.4, is illustrated at the frequency marked ω_1 . Two curves are drawn representing two of the elements that enter into the sum given in Eq. (5.9). The value of the larger amplitude component located at the frequency ω_1 is the value of $R(j\omega_1)$. The smaller

Figure 5.4

(a) Sketch of a spectrum amplitude and (b) the components of the spectrum after sampling, showing aliasing



component shown at ω_1 comes from the spectrum centered at $2\pi/T$ and is $R(j\omega_0)$, where ω_0 is such that $\omega_0 = \omega_1 - 2\pi/T$. This signal at frequency ω_0 which produces a component at frequency ω_1 after sampling is called in the trade an “alias” of ω_1 ; the phenomenon is called **aliasing**.

aliasing

The phenomenon of aliasing has a clear meaning in time. Two continuous sinusoids of different frequencies appear at the same frequency when sampled. We cannot, therefore, distinguish between them based on their samples alone. Figure 5.5 shows a plot of a sinusoid at $\frac{1}{8}$ Hz and of a sinusoid at $\frac{7}{8}$ Hz. If we sample these waves at 1 Hz, as indicated by the dots, then we get the same sample values from both signals and would continue to get the same sample values for all time. Note that the sampling frequency is 1, and, if $f_1 = \frac{1}{8}$, then

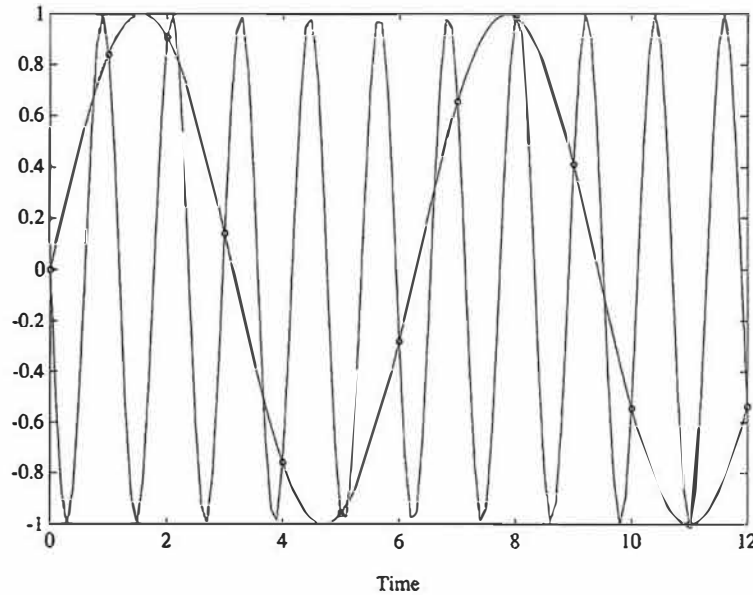
$$f_0 = \frac{1}{8} - 1 = -\frac{7}{8}.$$

The significance of the negative frequency is that the $\frac{7}{8}$ -Hz sinusoid in Fig. 5.5 is a negative sine function.

Thus, as a direct result of the sampling operation, when data are sampled at frequency $2\pi/T$, the total harmonic content at a given frequency ω_1 is to be found not only from the original signal at ω_1 but also from all those frequencies that are aliases of ω_1 , namely, components from all frequencies $\omega_1 + n2\pi/T = \omega_1 + n\omega_s$, as shown in the formula of Eq. (5.9) and sketched in Fig. 5.4. The errors caused by aliasing can be very severe if a substantial quantity of high-frequency components is contained in the signal to be sampled. To minimize the error caused by this

Figure 5.5

Plot of two sinusoids that have identical values at unit sampling intervals—an example of aliasing



effect, it is standard practice to *precede* the sampling operation (such as the sample-and-hold circuit discussed earlier) by a low-pass antialias filter that will remove substantially all spectral content above the half-sampling frequency, i.e., above π/T . A sketch suggesting the result of an anti-aliasing filter is drawn in Fig. 5.6.

sampling theorem

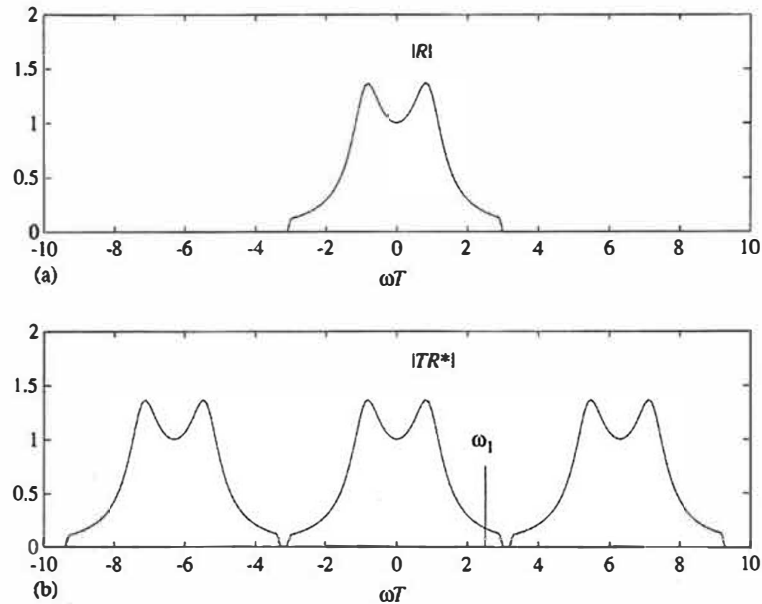
If all spectral content above the frequency π/T is removed, then no aliasing is introduced by sampling and the signal spectrum is not distorted, even though it is repeated endlessly, centered at $n2\pi/T$. The critical frequency, π/T , was first reported by H. Nyquist and is called the Nyquist frequency. Band-limited signals that have no components above the Nyquist frequency are represented unambiguously by their samples. A corollary to the aliasing issue is the **sampling theorem**. We have seen that if $R(j\omega)$ has components above the Nyquist frequency $\omega_s/2$ or π/T , then overlap and aliasing will occur. Conversely, we noticed that if $R(j\omega)$ is zero for $|\omega| \geq \pi/T$, then sampling at intervals of T sec. will produce no aliasing and the original spectrum can be recovered exactly from R^* , the spectrum of the samples. Once the spectrum is recovered by inverse transform, we can calculate the original signal itself. This is the sampling theorem: One can recover a signal from its samples if the sampling frequency ($\omega_s = 2\pi/T$) is *at least* twice the highest frequency (π/T) in the signal. Notice that the sampling theorem requires that $R(j\omega)$ is exactly zero for all frequencies above π/T .

hidden oscillations

A phenomenon somewhat related to aliasing is that of **hidden oscillations**. There is the possibility that a signal could contain some frequencies that the

Figure 5.6

(a) Sketch of a spectrum amplitude and (b) the components of the spectrum after sampling, showing removal of aliasing with an antialiasing filter



samples do not show *at all*. Such signals, when they occur in a digital control system, are called “hidden oscillations,” an example of which is shown in a design problem in Fig. 7.29. Hidden oscillations can only occur at multiples of the Nyquist frequency (π/T).

5.3 Data Extrapolation

The sampling theorem states that under the right conditions it is possible to recover a signal from its samples; we now consider a formula for doing so. From Fig. 5.6 we can see that the spectrum of $R(j\omega)$ is contained in the low-frequency part of $R^*(j\omega)$. Therefore, to recover $R(j\omega)$ we need only process $R^*(j\omega)$ through a low-pass filter and multiply by T . As a matter of fact, if $R(j\omega)$ has zero energy for frequencies in the bands above π/T (such an R is said to be band-limited), then an ideal low-pass filter with gain T for $-\pi/T \leq \omega \leq \pi/T$ and zero elsewhere would recover $R(j\omega)$ from $R^*(j\omega)$ exactly. Suppose we define this ideal low-pass filter characteristic as $L(j\omega)$. Then we have the result

$$R(j\omega) = L(j\omega)R^*(j\omega). \quad (5.10)$$

The signal $r(t)$ is the inverse transform of $R(j\omega)$, and because by Eq. (5.10) $R(j\omega)$ is the *product* of two transforms, its inverse transform $r(t)$ must be the convolution of the time functions $\ell(t)$ and $r^*(t)$. The form of the filter impulse

response can be computed by using the definition of $L(j\omega)$ from which the inverse transform gives

$$\begin{aligned}
 \ell(t) &= \frac{1}{2\pi} \int_{-\pi/T}^{\pi/T} T e^{j\omega t} d\omega \\
 &= \frac{T}{2\pi} \frac{e^{j\omega t}}{jt} \Big|_{-\pi/T}^{\pi/T} \\
 &= \frac{T}{2\pi jt} (e^{j(\pi t/T)} - e^{-j(\pi t/T)}) \\
 &= \frac{\sin(\pi t/T)}{\pi t/T} \\
 &\triangleq \text{sinc} \frac{\pi t}{T}.
 \end{aligned} \tag{5.11}$$

Using Eq. (5.1) for $r^*(t)$ and Eq. (5.11) for $\ell(t)$, we find that their convolution is

$$r(t) = \int_{-\infty}^{\infty} r(\tau) \sum_{k=-\infty}^{\infty} \delta(\tau - kT) \text{sinc} \frac{\pi(t - \tau)}{T} d\tau.$$

Using the sifting property of the impulse, we have

$$r(t) = \sum_{k=-\infty}^{\infty} r(kT) \text{sinc} \frac{\pi(t - kT)}{T}. \tag{5.12}$$

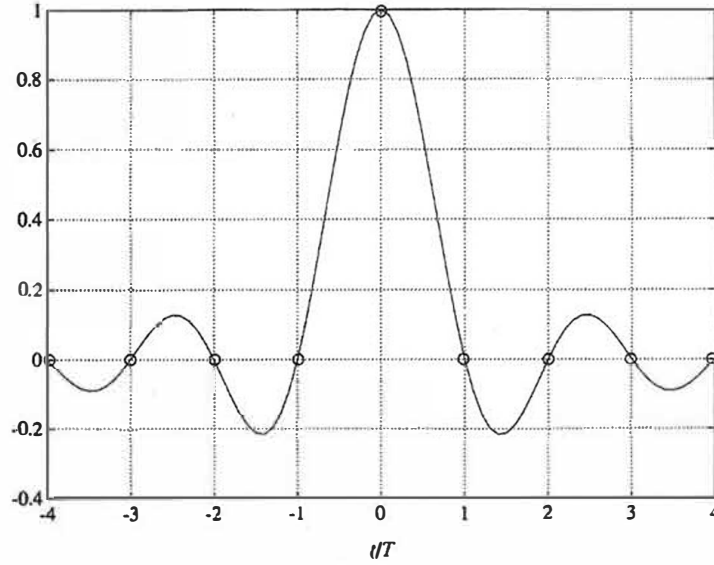
Equation (5.12) is a constructive statement of the sampling theorem. It shows explicitly how to construct a band-limited function $r(t)$ from its samples. The sinc functions are the interpolators that fill in the time gaps between samples with a signal that has no frequencies above π/T . A plot of the impulse response of this "ideal" hold filter is drawn in Fig. 5.7 from the formula of Eq. (5.11).

There is one serious drawback to the extrapolating signal given by Eq. (5.11). Because $\ell(t)$ is the impulse response of the ideal low-pass filter $L(j\omega)$, it follows that this filter is *noncausal* because $\ell(t)$ is nonzero for $t < 0$. $\ell(t)$ starts at $t = -\infty$ while the impulse that triggers it does not occur until $t = 0$! In many communications problems the interpolated signal is not needed until well after the samples are acquired, and the noncausality can be overcome by adding a phase lag, $e^{-j\omega\lambda}$, to $L(j\omega)$, which adds a *delay* to the filter and to the signals processed through it. In feedback control systems, a large delay is usually disastrous for stability, so we avoid such approximations to this function and use something else, like the polynomial holds, of which the zero-order hold already mentioned in connection with the ADC is the most elementary and the most common.

In Section 5.2 we introduced the zero-order hold as a model for the storage register in an A/D converter that maintains a constant signal value between samples. We showed in Eq. (5.7) that it has the transfer function

$$\text{ZOH}(j\omega) = \frac{1 - e^{-j\omega T}}{j\omega}. \tag{5.13}$$

Figure 5.7
Plot of the impulse
response of the ideal
low-pass filter



We can discover the frequency properties of $ZOH(j\omega)$ by expressing Eq. (5.13) in magnitude and phase form. To do this, we factor out $e^{-j\omega T/2}$ and multiply and divide by $2j$ to write the transfer function in the form

$$ZOH(j\omega) = e^{-j\omega T/2} \left\{ \frac{e^{j\omega T/2} - e^{-j\omega T/2}}{2j} \right\} \frac{2j}{j\omega}.$$

The term in brackets is recognized as the sine, so this can be written

$$ZOH(j\omega) = T e^{-j\omega T/2} \frac{\sin(\omega T/2)}{\omega T/2}$$

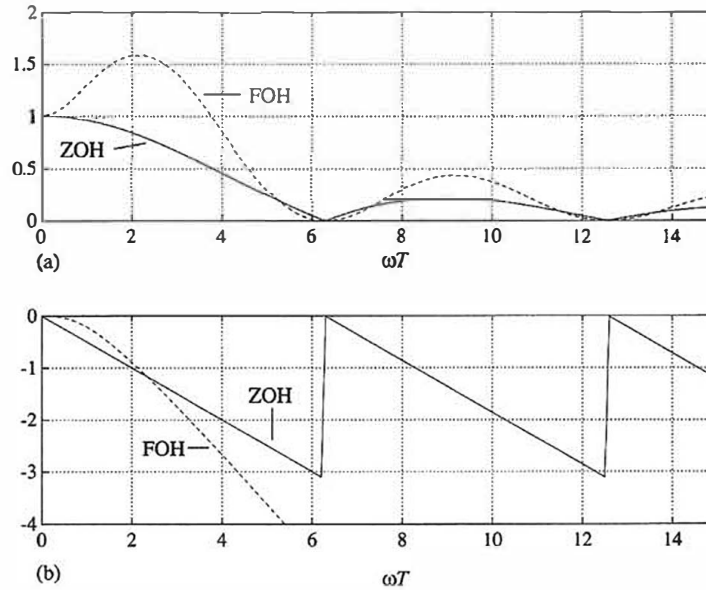
and, using the definition of the sinc function,

$$ZOH(j\omega) = e^{-j\omega T/2} T \text{sinc}(\omega T/2). \quad (5.14)$$

Thus the effect of the zero-order hold is to introduce a phase shift of $\omega T/2$, which corresponds to a time delay of $T/2$ seconds, and to multiply the gain by a function with the magnitude of $\text{sinc}(\omega T/2)$. A plot of the magnitude is shown in Fig. 5.8, which illustrates the fact that although the zero-order hold is a low-pass filter, it has a cut-off frequency well beyond the Nyquist frequency. The magnitude function is

$$|ZOH(j\omega)| = T \left| \text{sinc} \frac{\omega T}{2} \right|, \quad (5.15)$$

Figure 5.8
(a) Magnitude and (b)
phase of polynomial
hold filters



which slowly gets smaller as ω gets larger until it is zero for the first time at $\omega = \omega_s = 2\pi/T$. The phase is

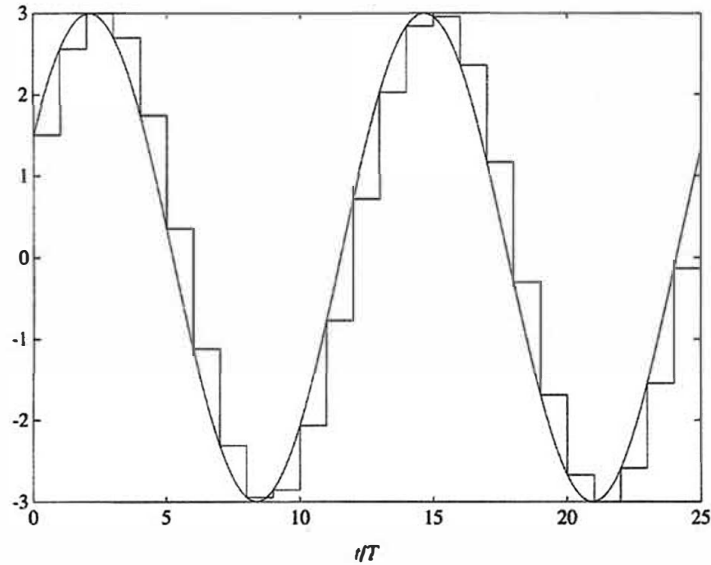
$$\angle ZOH(j\omega) = \frac{-\omega T}{2}, \quad (5.16)$$

plus the 180° shifts where the sinc function changes sign.

We can now give a complete analysis of the sample-and-hold circuit of Fig. 5.3(d) for a sinusoidal input $r(t)$ in both the time and the frequency domains. We consider first the time domain, which is simpler, being just an exercise in construction. For purposes of illustration, we will use $r(t) = 3 \sin(50t + \pi/6)$ as plotted in Fig. 5.9. If we sample $r(t)$ at the instants kT where the sampling frequency is $\omega_s = 2\pi/T = 200\pi$ and $T = 0.01$, then the plot of the resulting $r_h(kT)$ is as shown in Fig. 5.9. Notice that although the input is a single sinusoid, the output is clearly *not* sinusoidal. Thus it is not possible to describe this system by a transfer function, because the fundamental property of linear, time-invariant systems is that a sinusoid input produces an output that is a sinusoid of the same frequency and the relative amplitudes and phases determine the transfer function. The sample-and-hold system is linear but time varying. In the frequency domain, it is clear that the output $r_h(t)$ contains more than one frequency, and a complete analysis requires that we compute the amplitudes and phases of them all. However, in the application to control systems, the output of the hold will typically be applied to a dynamical system that is of low-pass character; thus the most important component in $r_h(t)$ is the fundamental harmonic, at $\omega_o = 50$

Figure 5.9

Plot of $3 \sin(50t + \pi/6)$
and the output of a
sample-and-hold with
sample period $T = 0.01$



rad/sec in this case. The other harmonics are *impostors*, appearing as part of the output signal when they are really unwanted consequences of the sample-and-hold process. In any event, we can proceed to analyze $r_h(t)$ for all its harmonics and select out the fundamental component, either by analysis or by a low-pass smoothing filter.

First, we need the spectrum of $r(t)$. Because a sinusoid can be decomposed into two exponentials, the spectrum of $r(t) = A \cos(\omega_o t + \phi)$ is given by two impulse functions at ω_o and $-\omega_o$ of intensity πA and phase ϕ and $-\phi$ as⁴

$$R(j\omega) = \pi A [e^{j\phi} \delta(\omega - \omega_o) + e^{-j\phi} \delta(\omega + \omega_o)].$$

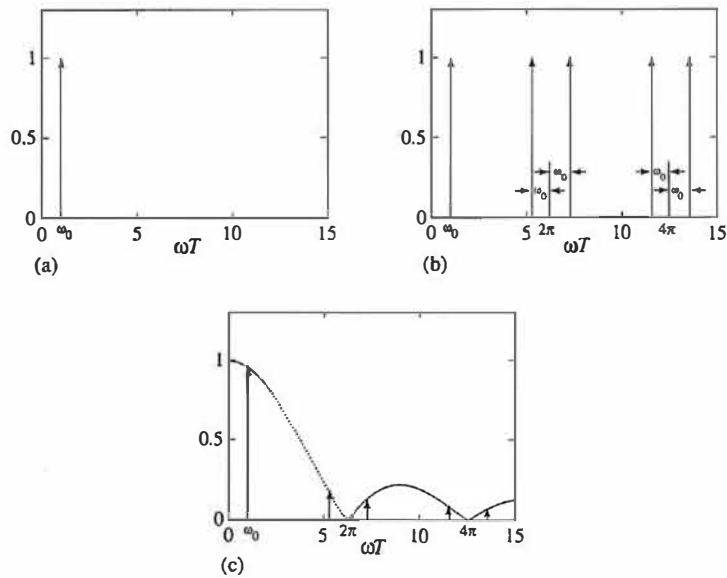
A sketch of this spectrum is shown in Fig 5.10(a) for $A = 1/\pi$. We represent the impulses by arrows whose heights are proportional to the intensities of the impulses.

After sampling, as we saw in Eq. (5.9), the spectrum of R^* is directly derived from that of R as the sum of multiple copies of that of R shifted by $n2\pi/T$ for all integers n and multiplied by $1/T$. A plot of the result normalized by T is shown in Fig. 5.10(b). Finally, to find the spectrum of R_h , we need only multiply the spectrum of R^* by the transfer function $ZOH(j\omega)$, which is

$$ZOH(j\omega) = T e^{-j\omega T/2} \text{sinc}(\omega T/2).$$

⁴ See the appendix to this chapter for details.

Figure 5.10
Plot of the spectra of
(a) R ; (b) R^* ; and (c) R_h



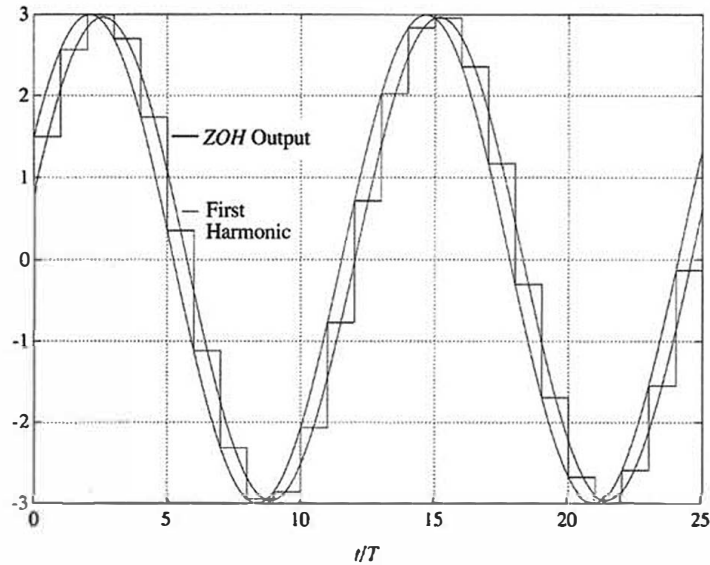
Thus the spectrum of R_h is also a sum of an infinite number of terms, but now with intensities modified by the sinc function and phases shifted by the delay function $\omega T/2$. These intensities are plotted in Fig. 5.10(c). Naturally, when all the harmonics included in R_h are converted to their time functions and added, they sum to the piecewise-constant staircase function plotted earlier in Fig. 5.9.

If we want a best approximation to r_h using only one sinusoid, we need only take out the first or fundamental harmonic from the components of R^* . This component has phase shift ϕ and amplitude $A \text{sinc}(\omega T/2)$. In the time domain, the corresponding sinusoid is given by

$$v_1(t) = A[\text{sinc}(\omega T/2)]\sin[\omega_\phi(t - \frac{T}{2})]. \quad (5.17)$$

A plot of this approximation for the signal from Fig 5.9 is given in Fig. 5.11 along with both the original input and the sampled-and-held output to show the nature of the approximation. In control design, we can frequently achieve a satisfactory design for a sampled-data system by approximating the sample and hold with a continuous transfer function corresponding to the delay of $T/2$. The controller design is then done in the continuous domain but is implemented by computing a discrete equivalent. More discussion of this technique, sometimes called *emulation*, will be given in Chapter 6, where some examples illustrate the results.

Figure 5.11
Plot of the output of the
sample and hold and the
first harmonic
approximation



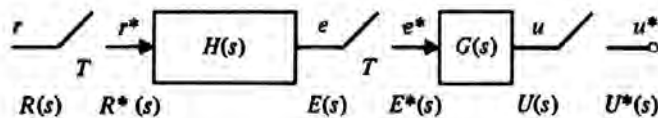
5.4 Block-Diagram Analysis of Sampled-Data Systems

We have thus far talked mainly about discrete, continuous, and sampled signals. To analyze a feedback system that contains a digital computer, we need to be able to compute the transforms of output signals of systems that contain sampling operations in various places, including feedback loops, in the block diagram. The technique for doing this is a simple extension of the ideas of block-diagram analysis of systems that are all continuous or all discrete, but one or two rules need to be carefully observed to assure success. First, we should review the facts of sampled-signal analysis.

We represent the process of sampling a continuous signal and holding it by impulse modulation followed by low-pass filtering. For example, the system of Fig. 5.12 leads to

$$\begin{aligned} E(s) &= R^*(s)H(s), \\ U(s) &= E^*(s)G(s). \end{aligned} \quad (5.18)$$

Figure 5.12
A cascade of samplers
and filters



Impulse modulation of continuous-time signals like $e(t)$ and $u(t)$ produces a series of sidebands as given in Eq. (5.9) and plotted in Fig. 5.4, which result in periodic functions of frequency. If the transform of the signal to be sampled is a product of a transform that is already periodic of period $2\pi/T$, and one that is not, as in $U(s) = E^*(s)G(s)$, where $E^*(s)$ is periodic and $G(s)$ is not, we can show that $E^*(s)$ comes out as a factor of the result. This is the most important relation for the block-diagram analysis of sampled-data systems, namely⁵

$$U^*(s) = (E^*(s)G(s))^* = E^*(s)G^*(s). \quad (5.19)$$

We can prove Eq. (5.19) either in the frequency domain, using Eq. (5.9), or in the time domain, using Eq. (5.1) and convolution. We will use Eq. (5.9) here. If $U(s) = E^*(s)G(s)$, then by definition we have

$$U^*(s) = \frac{1}{T} \sum_{n=-\infty}^{\infty} E^*(s - jn\omega_s)G(s - jn\omega_s); \quad (5.20)$$

but $E^*(s)$ is

$$E^*(s) = \frac{1}{T} \sum_{k=-\infty}^{\infty} E(s - jk\omega_s),$$

so that

$$E^*(s - jn\omega_s) = \frac{1}{T} \sum_{k=-\infty}^{\infty} E(s - jk\omega_s - jn\omega_s). \quad (5.21)$$

Now in Eq. (5.21) we can let $k = \ell - n$ to get

$$\begin{aligned} E^*(s - jn\omega_s) &= \frac{1}{T} \sum_{\ell=-\infty}^{\infty} E(s - j\ell\omega_s) \\ &= E^*(s). \end{aligned} \quad (5.22)$$

In other words, because E^* is already periodic, shifting it an integral number of periods leaves it unchanged. Substituting Eq. (5.22) into Eq. (5.20) yields

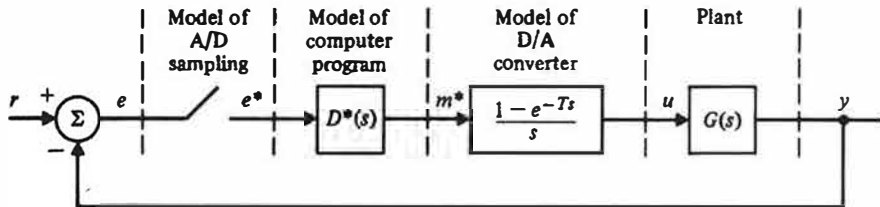
$$\begin{aligned} U^*(s) &= E^*(s) \frac{1}{T} \sum_{-\infty}^{\infty} G(s - jn\omega_s) \\ &= E^*(s)G^*(s). \quad \text{QED} \end{aligned} \quad (5.23)$$

Note especially what is *not* true. If $U(s) = E(s)G(s)$, then $U^*(s) \neq E^*(s)G^*(s)$ but rather $U^*(s) = (EG)^*(s)$. The periodic character of E^* in Eq. (5.19) is crucial.

⁵ We of course assume the existence of $U^*(s)$, which is assured if $G(s)$ tends to zero as s tends to infinity at least as fast as $1/s$. We must be careful to avoid impulse modulation of impulses, for $\delta(t)\delta(t)$ is undefined.

Figure 5.13

Block diagram of digital control as a sampled-data system



The final result we require is that, given a sampled-signal transform such as $U^*(s)$, we can find the corresponding z -transform simply by letting $e^{sT} = z$ or

$$U(z) = U^*(s) \big|_{e^{sT}=z}. \quad (5.24)$$

There is an important time-domain reflection of Eq. (5.24). The inverse Laplace transform of $U^*(s)$ is the sequence of *impulses* with intensities given by $u(kT)$; the inverse z -transform of $U(z)$ is the sequence of values $u(kT)$. Conceptually, sequences of values and the corresponding z -transforms are easy to think about as being processed by a computer program, whereas the model of sampling as a sequence of impulses is what allows us to analyze a discrete system embedded in a continuous world (see Fig. 5.13). Of course, the impulse modulator must *always* be eventually followed by a low-pass circuit (hold circuit) in the physical world. Note that Eq. (5.24) can also be used in the other direction to obtain $U^*(s)$, the Laplace transform of the train of impulses, from a given $U(z)$.

◆ Example 5.1 Block Diagram Analysis

Compute the transforms of Y^* and Y for the system block diagram of Fig. 5.13.

Solution. In Fig. 5.13 we have modeled the A/D converter plus computer program plus D/A converter as an impulse modulator [which takes the samples from $e(t)$], a computer program that processes these samples described by $D^*(s)$, and a zero-order hold that constructs the piecewise-constant output of the D/A converter from the impulses of m^* . In the actual computer we assume that the samples of $e(t)$ are manipulated by a difference equation whose input-output effect is described by the z -transform $D(z)$. These operations are represented in Fig. 5.13 as if they were performed on impulses, and hence the transfer function is $D^*(s)$ according to Eq. (5.24). Finally, the manipulated impulses, $m^*(t)$, are applied to the zero-order hold from which the piecewise-constant-control signal $u(t)$ comes. In reality, of course, the computer operates on the sample values of $e(t)$ and the piecewise-constant output is generated via a storage register and a D/A converter. The impulses provide us with a convenient, consistent,

and effective model of the processes to which Laplace-transform methods can be applied. From the results given thus far, we can write relations among Laplace transforms as

$$\begin{aligned} E(s) &= R - Y, & (a) \\ M^*(s) &= E^* D^*, & (b) \\ U &= M^* \left[\frac{1 - e^{-Ts}}{s} \right], & (c) \\ Y &= GU. & (d) \end{aligned} \quad (5.25)$$

The usual idea is to relate the discrete output, Y^* , to the discrete input, R^* . Suppose we sample each of these equations by using the results of Eq. (5.19) to "star" each transform. The equations are ⁶

$$\begin{aligned} E^* &= R^* - Y^*, & (a) \\ M^* &= E^* D^*, & (b) \\ U^* &= M^*, & (c) \\ Y^* &= [GU]^*. & (d) \end{aligned} \quad (5.26)$$

Now Eq. (5.26(d)) indicates that we need U , not U^* , to compute Y^* , so we must back up to substitute Eq. (5.25(c)) into Eq. (5.26(d)):

$$Y^* = \left[GM^* \left(\frac{1 - e^{-Ts}}{s} \right) \right]^*. \quad (5.27)$$

Taking out the periodic parts, which are those in which s appears only as e^{sT} [which include $M^*(s)$], we have

$$Y^* = (1 - e^{-Ts}) M^* \left(\frac{G}{s} \right)^*. \quad (5.28)$$

Substituting from Eq. (5.26(b)) for M^* gives

$$Y^* = (1 - e^{-Ts}) E^* D^* (G/s)^*. \quad (5.29)$$

And substituting Eq. (5.26(a)) for E^* yields

$$Y^* = (1 - e^{-Ts}) D^* (G/s)^* [R^* - Y^*]. \quad (5.30)$$

If we call

$$(1 - e^{-Ts}) D^* (G/s)^* = H^*, \quad (5.31)$$

then we can solve Eq. (5.30) for Y^* , obtaining

$$Y^* = \frac{H^*}{1 + H^*} R^*. \quad (5.32)$$

⁶ In sampling Eq. (5.25(c)) we obtain Eq. (5.26(c)) by use of the continuous-from-the-right convention for Eq. (5.5) for impulse modulation of discontinuous functions. From the time-domain operation of the zero-order hold, it is clear that the samples of u and m are the same, and then from this Eq. (5.26(c)) follows.

◆ **Example 5.2** Analysis of a Specific Block Diagram

Apply the results of Example 1 to compute Y^* and Y for the case where

$$G(s) = \frac{a}{s+a}, \quad (5.33)$$

and the sampling period T is such that $e^{-aT} = \frac{1}{2}$. The computer program corresponds to a discrete integrator

$$u(kT) = u(kT - T) + K_0 e(kT), \quad (5.34)$$

and the computer D/A holds the output constant so that the zero-order hold is the correct model.

Solution. We wish to compute the components of H^* given in Eq. (5.31). For the computer program we have the transfer function of Eq. (5.34), which in terms of z is

$$D(z) = \frac{U(z)}{E(z)} = \frac{K_0}{1-z^{-1}} = \frac{K_0 z}{z-1}.$$

Using Eq. (5.24), we get the Laplace-transform form

$$D^*(s) = \frac{K_0 e^{sT}}{e^{sT} - 1}. \quad (5.35)$$

For the plant and zero-order-hold we require

$$\begin{aligned} (1 - e^{-Ts})(G(s)/s)^* &= (1 - e^{-Ts}) \left(\frac{a}{s(s+a)} \right)^* \\ &= (1 - e^{-Ts}) \left(\frac{1}{s} - \frac{1}{s+a} \right)^* \end{aligned}$$

Using Eq. (5.5), we have

$$(1 - e^{-Ts})(G(s)/s)^* = (1 - e^{-Ts}) \left(\frac{1}{1 - e^{-Ts}} - \frac{1}{1 - e^{-aT} e^{-Ts}} \right).$$

Because $e^{-aT} = \frac{1}{2}$, this reduces to

$$\begin{aligned} (1 - e^{-Ts})(G(s)/s)^* &= \frac{(1/2)e^{-Ts}}{1 - (1/2)e^{-Ts}} \\ &= \frac{1/2}{e^{Ts} - 1/2}. \end{aligned} \quad (5.36)$$

Combining Eq. (5.36) and Eq. (5.35) then, in this case, we obtain

$$H^*(s) = \frac{K_0}{2} \frac{e^{sT}}{(e^{sT} - 1)(e^{sT} - 1/2)}. \quad (5.37)$$

Equation (5.37) can now be used in Eq. (5.32) to find the closed-loop transfer function from which the dynamic and static responses can be studied, as a function of K_0 , the program gain. We note also that beginning with Eq. (5.25), we can readily calculate that

$$Y(s) = R^* \frac{D^*}{1 + H^*} \frac{(1 - e^{-Ts})}{s} G(s). \quad (5.38)$$



Equation (5.38) shows how to compute the response of this system in between sampling instants. For a given $r(t)$, the starred terms in Eq. (5.38) and the $(1 - e^{-Ts})$ -term correspond to a train of impulses whose individual values can be computed by expanding in powers of e^{-Ts} . These impulses are applied to $G(s)/s$, which is the step response of the plant. Thus, between sampling instants, we will see segments of the plant step response.

With the exception of the odd-looking forward transfer function, Eq. (5.32) looks like the familiar feedback formula: forward-over-one-plus-feedback. Unfortunately, the sequence of equations by which Eq. (5.32) was computed was a bit haphazard, and such an effort might not always succeed. Another example will further illustrate the problem.

◆ Example 5.3 Another Block Diagram Analysis

Compute Y^* and Y for the block diagram of Fig. 5.14.

Solution. The equations describing the system are (all symbols are Laplace transforms)

$$\begin{aligned} E &= R - Y, & (a) \\ U &= HE, & (b) \\ Y &= U^*G; & (c) \end{aligned} \tag{5.39}$$

and after sampling, the equations are

$$\begin{aligned} E^* &= R^* - Y^*, & (a) \\ U^* &= (HE)^*, & (b) \\ Y^* &= U^*G^*. & (c) \end{aligned} \tag{5.40}$$

How do we solve these? In Eq. (5.40(b)) we need E , not E^* . So we must go back to Eq. (5.39(a))

$$\begin{aligned} U^* &= (H(R - Y))^* \\ &= (HR)^* - (HY)^*. \end{aligned}$$

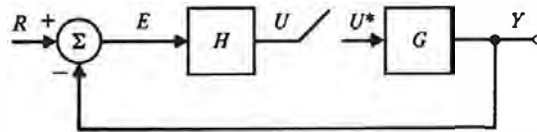
Using Eq. (5.39(c)) for Y , we have

$$U^* = (HR)^* - (HU^*G)^*.$$

Taking out the periodic U^* in the second term on the right gives

$$U^* = (HR)^* - U^*(HG)^*.$$

Figure 5.14
A simple system that does not have a transfer function



Solving, we get

$$U^* = \frac{(HR)^*}{1 + (HG)^*}. \quad (5.41)$$

From Eq. (5.40(c)), we can solve for Y^*

$$Y^* = \frac{(HR)^*}{1 + (HG)^*} G^*. \quad (5.42)$$

Equation (5.42) displays a curious fact. The transform of the input is bound up with $H(s)$ and *cannot* be divided out to give a transfer function! This system displays an important fact that with the manipulations of stars for sampling might be overlooked: A sampled-data system is *time varying*. The response depends on the time *relative to the sampling instants* at which the signal is applied. Only when the input samples *alone* are required to generate the output samples can we obtain a discrete transfer function. The time variation occurs on the taking of samples. In general, as in Fig. 5.14, the entire input signal $r(t)$ is involved in the system response, and the transfer-function concept fails. Even in the absence of a transfer function, however, the techniques developed here permit study of stability and response to specific inputs such as step, ramp, or sinusoidal signals.

We need to know the general rules of block-diagram analysis. In solving Fig. 5.14 we found ourselves working with U , the signal that was sampled. This is in fact the key to the problem. Given a block diagram with several samplers, *always select the variables at the inputs to the samplers as the unknowns*. Being sampled, these variables have periodic transforms that will always factor out after the sampling process and result in a set of equations in the sampled (starred) variables that can be solved.

◆ Example 5.4 Another Block Diagram Analysis

Compute the transforms of Y^* and Y for the block diagram drawn in Fig. 5.15.

Solution. We select E and M as independent variables and write

$$\begin{aligned} E(s) &= R - M^* G_2, \\ M(s) &= E^* H G_1. \end{aligned} \quad (5.43)$$

Figure 5.15
A final example for
transfer-function analysis
of sampled-data systems

